

Solution Architecture

Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Team ID	SWTID-2026-7239
Team Lead	Divyadharshini R
Team Members	Hithisri V, Harini A, Manju R, Preethi V

Architecture Overview

ComicCraft is organized into three primary components: Frontend, Backend, and AI Integration.

Frontend

HTML, CSS, and Jinja2 templates. index.html collects inputs; comic_preview.html displays the generated comic; export_success.html confirms successful export.

Backend

FastAPI handles routing, input processing, AI workflow orchestration, error handling, layout creation, and PDF export.

AI Integration

Gemini Flash creates the structured 5-panel outline; Gemini Pro creates detailed narration/dialogues; Stable Diffusion creates comic-style images.

Core Modules

gemini_flash.py: generate_outline(); gemini_pro.py: generate_story(); image_generator.py: generate_image(); layout_builder.py: build_comic_layout(); exporters.py: save_pdf(); routes.py: application routes.

Storage Paths

Generated panel images are saved under static/panels; exported PDF files are saved under static/exports.

Architecture Flow

Browser → FastAPI routes.py → Gemini/Stable Diffusion modules → layout builder → PDF exporter → Jinja2 preview/export pages.